Maathrucare: An AI-Based Pregnancy Support System for Sri Lankan Mothers

INDIVIDUAL THESIS

RANDINU P.V.K.V

(IT22557506)

Individual component: Pregnancy Risk Prediction & Health Monitoring Module

(Component 2 of Maathru Care)

Group ID: R26-IT-146

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration: I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure: I have disclosed all use of generative AI and other automated tools in accordance with the module policy. I remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this thesis.

Student name: RANDINU P.V.K.V Student ID: IT22557506

Signature:  Date: 10/11/2026

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Dr. Kapila Dissanayake		
Co-supervisor: Mr. Gayan		

ABSTRACT

Maternal mortality remains a significant global health challenge, with a substantial proportion of pregnancy-related deaths attributable to preventable complications that go undetected between clinical visits. This individual component addresses that gap within the Maathru Care maternal health monitoring system by designing, implementing, and evaluating a Pregnancy Risk Prediction and Health Monitoring module that uses a machine-learning classifier to assess a pregnant woman's risk level from thirteen clinical features collected through a mobile application. The component classifies risk into four tiers — Low, Moderate, High, and Very High — using a gradient-boosted tree classifier (XGBoost) trained on a labelled dataset of 998 maternal health records. Following systematic hyperparameter tuning using grid search with five-fold cross-validation, the tuned model achieved 98.00% test accuracy with a cross-validation score of 98.00% ± 0.71%, substantially outperforming Logistic Regression (79.00%) and approaching Random Forest (97.50%) while benefiting from XGBoost's built-in regularization. The component integrates real-time doctor alert notifications, emergency SOS functionality, risk trend analysis, daily symptom tracking, doctor-patient in-app chat, clinical note delivery, and appointment scheduling into a cohesive maternal health platform. SHAP (SHapley Additive exPlanations) values provide per-prediction transparency, identifying which clinical features most influenced each individual risk assessment. The component was implemented as a three-tier architecture comprising a React Native mobile application, a Node.js Express backend API gateway, and a Python FastAPI microservice hosting the trained XGBoost model, with a Supabase PostgreSQL cloud database. The findings demonstrate that an XGBoost-based, mobile-accessible risk classification system can achieve clinically meaningful prediction accuracy while maintaining transparent, explainable outputs suitable for an AI-assisted decision-support role in maternal healthcare.

Keywords: maternal health, pregnancy risk prediction, XGBoost, SHAP explainability, mobile health, React Native, FastAPI, alert classification, antenatal monitoring

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Dr. Kapila Dissanayake, for their guidance, constructive feedback, and encouragement throughout the design, implementation, and evaluation of this component. I am grateful to my fellow team members for their collaboration in integrating this module into the complete Maathru Care system and for cooperation in defining the shared API contracts and data schemas that span all four components. I also extend my appreciation to the Faculty of Computing at the Sri Lanka Institute of Information Technology for the academic resources, infrastructure, and support provided throughout this project.

TABLE OF CONTENTS

1 INTRODUCTION 10
1.1 Overall project context 10
1.2 Individual component context 10
1.3 Problem statement 10
1.4 Research gap 11
1.5 Aim and objectives 11
1.6 Research questions or hypotheses 11
1.7 Scope limitations and assumptions 12
1.8 Individual contribution 12
1.9 Thesis structure 12
2 LITERATURE REVIEW AND RELATED WORK 12
2.1 Review method and scope 12
2.2 Domain and theoretical foundations 13
2.3 Related research 13
2.4 Existing systems products and techniques 14
2.5 Gap synthesis and conceptual framework 14
3 RESEARCH METHODOLOGY 14
3.1 Research design 14
3.2 Data participants or study materials 15
3.3 Experimental or study procedure 15
3.4 Measures and evaluation metrics 15
3.5 Data analysis 15
3.6 Ethics privacy security and responsible AI 16
3.7 Reproducibility and research artefacts 16
4 PRODUCT AND SOLUTION DESIGN 16
4.1 Requirements 16
4.2 Component architecture within the complete solution 17
4.3 Design alternatives and decisions 19
4.4 Detailed design 20
4.5 Standards and professional practice 20
5 IMPLEMENTATION, INTEGRATION AND TESTING 21
5.1 Development environment and technology stack 21
5.2 Component implementation 21
5.3 Integration with the other components 22
5.4 Verification and software testing 22
5.4.1 Test case summary 23
5.5 Deployment and operation 23

6 RESULTS AND VALIDATION 23
6.1 Research results 23
6.2 Product and technical results 24
6.3 User or stakeholder validation 25
6.4 Integration results 25
6.5 Summary of findings 25
7 DISCUSSION 26
7.1 Interpretation of findings 26
7.2 Comparison with literature baselines and existing solutions 26
7.3 Research contribution 26
7.4 Product contribution 27
7.5 Limitations and threats to validity 27
7.6 Implications 27
8 COMMERCIALISATION AND SUSTAINABILITY 27
8.1 Value proposition and target users 27
8.2 Market and competitive position 28
8.3 Business or sustainability model 28
8.4 Scalability, intellectual property and compliance 28
8.5 Contribution to the Sustainable Development Goals 28
9 CONCLUSION AND FUTURE DIRECTIONS 29
9.1 Conclusion 29
9.2 Achievement of objectives 29
9.3 Individual research and product contributions 30
9.4 Future directions 30
REFERENCES 30
GLOSSARY 31
APPENDICES 32
Appendix A Research instruments or extended results 32
Appendix B Academic integrity and AI use disclosure 32

LIST OF FIGURES

Figure	Title
Figure 4.1	Component architecture of the Pregnancy Risk Prediction module within the Maathru Care solution
Figure 4.2	AI risk prediction data flow: mobile input to XGBoost inference to mobile output
Figure 6.1	Model accuracy comparison: Logistic Regression, Random Forest, and Tuned XGBoost
Figure 6.2	SHAP global feature importance ranking for the tuned XGBoost model

LIST OF TABLES

Table	Title
Table 4.1	Thirteen input features: name, type, value range, and clinical rationale
Table 5.1	Development environment and technology stack
Table 5.2	Functional and non-functional test case summary
Table 6.1	Classification report: precision, recall, and F1-score for the tuned XGBoost model
Table 6.2	Model comparison summary

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BP	Blood Pressure
CDSS	Clinical Decision Support System
CV	Cross-Validation
FHR	Fetal Heart Rate
GDM	Gestational Diabetes Mellitus
HBsAg	Hepatitis B Surface Antigen
HTTP	HyperText Transfer Protocol
ML	Machine Learning
REST	Representational State Transfer
SHAP	SHapley Additive exPlanations
SOS	Save Our Souls (Emergency Signal)
SQL	Structured Query Language
VDRL	Venereal Disease Research Laboratory
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall project context

Maathru Care is a four-component maternal health monitoring system developed as a final-year group project at the Sri Lanka Institute of Information Technology (SLIIT). The system targets the critical gap between scheduled antenatal clinic visits — during which a clinician physically assesses a pregnant woman's condition — and the weeks spent at home during which health status changes may go undetected. Maathru Care addresses this gap by providing continuous, AI-assisted health monitoring through a mobile application accessible to both pregnant mothers and their assigned obstetric physicians. The four components of the system are: a CTG waveform analysis module (Member 1), the Pregnancy Risk Prediction and Health Monitoring module described in this thesis (Member 2), a a CTG waveform-based fetal monitoring module (Member 3), and a a user management and approval workflow module (Member 4). The four components share a common cloud database (Supabase PostgreSQL) and communicate through defined REST API contracts documented in the group's Common Report.

1.2 Individual component context

This thesis describes the Pregnancy Risk Prediction and Health Monitoring module, which is responsible for the core AI-driven clinical decision-support functionality of the Maathru Care system. The module accepts thirteen clinical feature values entered by the pregnant mother through the mobile application, submits them to a trained XGBoost classifier hosted as a Python FastAPI microservice, and returns a structured risk assessment comprising a four-tier risk level (Low, Moderate, High, Very High), a high-risk probability percentage, per-feature SHAP explanations, and evidence-based preventive guidance. When the assessed risk is classified as High or Very High, the module automatically creates an alert record that notifies the patient's assigned doctor through the doctor-facing dashboard. Beyond the risk assessment itself, this component also implements the complete patient-facing and doctor-facing health management interfaces: daily symptom logging, BP and weight vitals tracking, medication and prescription management, in-app doctor-patient chat, appointment scheduling and notification, emergency SOS alert dispatch, and risk trend visualization.

1.3 Problem statement

Pregnancy-related complications — including pre-eclampsia, gestational diabetes, and foetal distress — are among the leading causes of maternal and perinatal mortality in low- and middle-income countries. A consistent pattern in maternal mortality audits is that many deaths were potentially avoidable had warning signs been identified and acted upon earlier. In current clinical practice in Sri Lanka and similar healthcare settings, a pregnant mother typically visits an antenatal clinic monthly or fortnightly; between visits, physiological changes that may signal developing complications go unmonitored unless the mother independently seeks care. No automated, personalized tool exists that integrates multiple clinical features from a mother's own routine records and provides her and her doctor with a continuously updated, AI-generated risk estimate — with an explanation of which features drove that estimate — between clinic visits.

1.4 Research gap

Existing mobile health applications for pregnancy monitoring typically provide educational content, appointment reminders, and symptom diaries, but do not integrate a trained, validated machine-learning classifier that computes a personalized risk probability from multiple clinical measurements simultaneously. The academic literature contains validated predictive models for specific conditions such as pre-eclampsia or gestational diabetes in isolation, but these models are rarely integrated into accessible mobile platforms with real-time doctor notification, per-prediction explainability, and full audit history. This component fills that gap by combining a high-accuracy XGBoost classifier, SHAP-based explainability, and a full-featured health monitoring mobile application into a unified, integrated module.

1.5 Aim and objectives

The aim of this component is to design, implement, and evaluate an AI-assisted maternal risk prediction and health monitoring module suitable for integration into the Maathru Care mobile platform.

The following five objectives were established:

- Objective 1 (feature selection and model design): Identify and encode thirteen clinically relevant maternal health features, and design a binary classification task distinguishing high-risk from non-high-risk pregnancies.
- Objective 2 (classifier training and tuning): Train an XGBoost classifier on a labelled maternal health dataset and systematically tune its hyperparameters to maximize cross-validation accuracy.
- Objective 3 (explainability): Integrate SHAP explainability so that each prediction is accompanied by a ranked, feature-level explanation, expressed in terms suitable for a non-specialist user.
- Objective 4 (integration): Integrate the trained classifier into the shared Maathru Care backend and deliver its outputs to the correct patient and doctor interfaces in real time.
- Objective 5 (engineering issue documentation): Identify, diagnose, resolve, and document any deployment-level issues encountered during integration, as reusable lessons for similar systems.

1.6 Research questions or hypotheses

- RQ1: Can a tuned XGBoost classifier trained on 998 maternal health records achieve a test accuracy of 95% or greater on a held-out test set?
- RQ2: Do SHAP global feature importance rankings for the trained model agree substantially with the feature importance rankings produced by an independently trained Random Forest model on the same dataset?
- RQ3: Can the trained classifier be integrated into a mobile health application with sufficiently low prediction latency that the risk result is returned to the user within a perceptibly short time after form submission?

1.7 Scope limitations and assumptions

This component is explicitly scoped as an AI-assisted decision-support tool, not a medical diagnostic system. The trained model was developed and evaluated on a dataset of 998 records and has not been clinically validated against prospective patient data. Risk level thresholds (Low: < 30%, Moderate: 30–59.99%, High: 60–79.99%, Very High: ≥ 80%) are application-defined thresholds for prototype demonstration purposes and are not derived from clinical guidelines. The component does not include CTG waveform analysis, which is the responsibility of Member 1. The component does not prescribe medication or autonomously make clinical decisions; all clinical decisions remain with the assigned physician. The component assumes that users are connected to the internet and that the backend servers are reachable; offline operation is not implemented in the current prototype.

1.8 Individual contribution

The candidate is solely responsible for all work described in this thesis: the XGBoost model training, hyperparameter tuning, and evaluation; the SHAP explainability integration; the Python FastAPI AI service; the Node.js Express backend routes for risk, alerts, vitals, prescriptions, doctor operations, and chat; the complete React Native mobile application screens for the mother-facing and doctor-facing interfaces; the Supabase database schema for all tables used by this component; and the emergency SOS and real-time doctor alert notification system. The candidate collaborated with the other three team members on defining the shared authentication contract, the database user table schema, and the inter-component API structure, as documented in the group's Common Report.

1.9 Thesis structure

The remainder of this thesis is organized as follows. Chapter 2 reviews related work in maternal risk prediction, mobile health applications, and explainable AI in healthcare. Chapter 3 describes the research methodology, including the Design Science Research approach and the dataset. Chapter 4 presents the product and solution design, including the architecture and the thirteen-feature representation. Chapter 5 describes implementation, integration, and testing. Chapter 6 reports evaluation results. Chapter 7 discusses findings, limitations, and implications. Chapter 8 considers commercialization and sustainability. Chapter 9 concludes the thesis and outlines future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This review covers four bodies of literature relevant to the Pregnancy Risk Prediction module: machine-learning approaches to maternal risk stratification; explainable AI methods suitable for healthcare classification tasks; mobile health (mHealth) application design for maternal and antenatal care; and evaluation-methodology practices for reporting classifier performance in clinical decision-support contexts. Sources were identified through targeted searches of IEEE Xplore, ACM Digital Library, PubMed, and general web search for peer-reviewed articles, conference proceedings, and systematic reviews, prioritizing sources that specifically addressed

prediction of pregnancy complications from structured clinical data, SHAP explainability for tree-based models, or the design of mobile health platforms for maternal monitoring.

2.2 Domain and theoretical foundations

Maternal mortality and pregnancy-related morbidity are recognized global public health priorities. The World Health Organization estimates that approximately 287,000 maternal deaths occurred globally in 2020, with the vast majority occurring in low- and middle-income countries. A consistent finding in maternal mortality audits is that a significant proportion of deaths were potentially avoidable through earlier detection of warning signs such as hypertension, abnormal fetal heart rate patterns, or glycosuria [1]. This clinical background motivates the inclusion of blood pressure measurements (systolic and diastolic BP), fetal heart rate, and urine sugar as primary features in the classification model described in this thesis, since these are routinely measured antenatal parameters with established associations with pre-eclampsia, fetal distress, and gestational diabetes respectively.

Antenatal care protocols in Sri Lanka and similar health systems rely on the Maternal Health Clinic Card, a standardized record maintained by each pregnant mother that documents the exact thirteen parameters used as features in this component: age, gravida, tetanus vaccination status, gestational age, weight, height, fetal position, fetal heart rate, urine sugar, VDRL result, HBsAg result, systolic BP, and diastolic BP. This protocol alignment means the feature set is directly derivable from existing clinical documentation rather than requiring additional instrumentation.

2.3 Related research

Machine-learning approaches to maternal risk stratification have been investigated in a number of recent studies. Saxena et al. compared Logistic Regression, Decision Tree, and Random Forest classifiers for predicting high-risk pregnancies from structured antenatal data, finding that ensemble tree methods consistently outperformed linear models, with Random Forest achieving accuracy in the range of 94–97% depending on feature selection [2]. This provides a baseline for comparison with the XGBoost model evaluated in Chapter 6 of this thesis. Ahmed et al. evaluated gradient-boosted classifiers for gestational diabetes prediction and reported that XGBoost outperformed both Random Forest and Support Vector Machine on a dataset of comparable size to the one used in this component, benefiting from its native handling of missing values and its regularization terms [3].

On explainability, Lundberg and Lee introduced SHAP as a unified framework for explaining the output of any machine-learning model by computing Shapley values — a game-theoretic concept measuring each feature's marginal contribution to a prediction [4]. SHAP TreeExplainer, the variant used in this component, computes exact Shapley values for tree-based models such as XGBoost in polynomial rather than exponential time, making it practical for per-prediction explanation in a real-time API context. The clinical applicability of SHAP for communicating model reasoning to non-specialist users has been demonstrated in several healthcare AI studies, though these studies also note the risk that SHAP explanations can be misinterpreted as causal if not carefully worded [5].

2.4 Existing systems products and techniques

Existing commercial maternal health mobile applications — including Ovia Pregnancy, What to Expect, and BabyCenter — provide rich educational content, symptom tracking diaries, and appointment reminders, but do not integrate a trained, personalized risk classifier that computes a patient-specific probability from clinical measurements. These applications treat the user primarily as a passive recipient of general information rather than as a source of clinical data to be actively analyzed. The gap between rich consumer health apps and clinical decision-support tools motivated the design of this component: a system that combines the accessibility of a consumer mobile app with the analytical capability of a validated ML classifier and the professional-facing alert infrastructure of a CDSS.

In the research literature, several prototype systems have combined mobile data collection with server-side risk classification for maternal health, including systems based on IoT wearable sensors for blood pressure monitoring [6]. However, these prototypes typically address a single condition rather than a multi-feature, multi-condition risk assessment, and rarely integrate the full patient-doctor workflow — alerts, notes, appointment scheduling, and chat — that the Maathru Care system implements.

2.5 Gap synthesis and conceptual framework

Synthesizing the literature above identifies a specific, addressable gap: a mobile-accessible, multi-feature maternal risk classification system that (a) integrates a validated ML classifier, (b) provides per-prediction SHAP explanations expressed in non-specialist language, (c) supports real-time doctor notification for high-risk assessments, and (d) incorporates the complete patient-doctor interaction workflow, has not been demonstrated in either the academic literature or commercial products in the Sri Lankan healthcare context. The conceptual framework adopted by this component treats maternal risk assessment as a supervised binary classification problem (High Risk vs Not High Risk) with a four-tier application-level risk display, using a gradient-boosted tree classifier whose decisions are explained using SHAP global and local feature importance, delivered through a three-tier mobile application architecture.

3 RESEARCH METHODOLOGY

3.1 Research design

This component follows a Design Science Research (DSR) methodology, as formalized by Hevner et al. [7], which is appropriate when the primary research output is a software artefact designed to address a practical problem, evaluated against measurable criteria. DSR is the appropriate methodology for this component because the deliverable is not a finding about the world but a designed artefact — the Pregnancy Risk Prediction and Health Monitoring module — that is evaluated for its technical performance and its contribution to solving the problem of gap-period maternal risk monitoring. The work progressed through problem identification (Chapter 1), objective definition (Section 1.5), literature review and gap identification (Chapter 2), artefact design (Chapter 4), implementation (Chapter 5), and evaluation (Chapter 6), with design revisions made in response to findings during development.

3.2 Data participants or study materials

The classifier was trained on a labelled dataset of 998 maternal health records, each containing the thirteen clinical features listed in Table 4.1 and a ground-truth binary label (HighRiskPregnancy: 1 = High Risk, 0 = Not High Risk). The dataset contains 666 High Risk records (66.7%) and 332 Not High Risk records (33.3%), representing an approximately 2:1 class imbalance. Before model training, the original patient-name field was excluded from features, and several highly incomplete or clinically uninformative fields identified during dataset inspection were also excluded. The retained thirteen features correspond directly to the parameters routinely recorded on the Sri Lankan Maternal Health Clinic Card, establishing face validity for the feature set. This dataset is a retrospective clinical dataset rather than a prospectively collected research dataset; its use is limited to the development and evaluation of a prototype decision-support tool, and no identifiable personal data was used in this thesis.

3.3 Experimental or study procedure

The dataset was partitioned into training and test sets using stratified sampling to preserve the 2:1 class ratio in both splits, yielding 798 training samples (80%) and 200 test samples (20%). Three classifier types were evaluated: Logistic Regression (as a linear baseline), Random Forest (as an ensemble baseline), and XGBoost (as the primary candidate). For XGBoost, systematic hyperparameter tuning was performed using GridSearchCV with five-fold cross-validation over a defined grid of learning rates, tree depths, and estimator counts, optimising accuracy. The best hyperparameter configuration — n_estimators=400, learning_rate=0.05, max_depth=3, random_state=42 — was selected based on the cross-validation score of 98.10%. The test set was held out completely from this search and used only for the final evaluation reported in Chapter 6. The trained model was serialised using joblib and loaded into a Python FastAPI service at application startup, so that prediction requests require inference only rather than retraining.

3.4 Measures and evaluation metrics

Overall model quality was measured using test-set accuracy and five-fold cross-validation accuracy with standard deviation, the latter serving as a measure of generalization stability. Per-class performance was measured using precision, recall, and F1-score for each of the two output classes (Not High Risk, High Risk), since accuracy alone can mask poor performance on one class in an imbalanced dataset. A confusion matrix was computed to characterize the distribution of true positives, true negatives, false positives, and false negatives. The Spearman rank correlation between SHAP global mean absolute importance rankings and Random Forest feature importance rankings was computed to address RQ2, testing the agreement between two independent importance measures.

3.5 Data analysis

Quantitative results were analyzed by comparing the three classifiers' accuracy, cross-validation score, and per-class precision/recall/F1-score (Chapter 6). The SHAP TreeExplainer was applied to the trained XGBoost model to compute both global (dataset-level) and local (per-prediction) SHAP values. Global importance was ranked by mean absolute SHAP value across

the training set, revealing WeightKg, TetanusVaccination, and DiastolicBP as the three most influential features. Local SHAP values were used to generate the per-prediction "Key Influencing Factors" explanations displayed in the mobile application.

3.6 Ethics privacy security and responsible AI

Because this component processes sensitive maternal health data — including blood pressure readings, laboratory results, and pregnancy history — its design treats all user data as health-adjacent and privacy-sensitive. The application handles this information under the following principles: data is stored in a cloud database with role-based access control enforced at the API layer; authentication is required for all API endpoints; environment variables are used for all secrets and API keys; and the mobile application does not cache sensitive health data in plaintext local storage. From a responsible AI perspective, the component's outputs are explicitly framed as AI-generated estimates for decision-support purposes, not medical diagnoses. The SHAP explanations presented to users are worded as "contributed to the model prediction" rather than "causes" to avoid implying medical causation. A disclaimer is displayed on every risk result screen: "This AI-generated assessment is for decision-support purposes and does not replace professional medical evaluation."

3.7 Reproducibility and research artefacts

The trained model (pregnancy_risk_xgboost.pkl), the Jupyter notebook containing the full training, evaluation, and SHAP analysis pipeline (01_dataset_inspection.ipynb), the Python FastAPI service, and the Node.js backend source code are all retained as project artefacts alongside the training dataset. The reported test accuracy of 98.00% and cross-validation score of 98.00% ± 0.71% can be reproduced by re-running the training notebook with random_state=42 on the same dataset.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The following functional and non-functional requirements were established for this component, derived from the problem statement in Section 1.3 and the research objectives in Section 1.5.

Functional Requirements:

- FR1: The component shall accept thirteen clinical feature values submitted through the mobile application and return a risk level (Low / Moderate / High / Very High), a high-risk probability percentage, and per-feature SHAP explanations.
- FR2: The component shall automatically create a doctor alert when the returned risk level is High or Very High, and make that alert visible on the assigned doctor's dashboard.
- FR3: The component shall support an emergency SOS function that allows a pregnant mother to dispatch an immediate alert to her assigned doctor with a full-screen, vibrating notification on the doctor's device.

- FR4: The component shall maintain a persistent risk assessment history for each patient, accessible as a trend graph and a comparative "Why did my risk change?" analysis.
- FR5: The component shall support daily symptom logging, BP and weight vitals tracking, and medication and prescription viewing.
- FR6: The component shall provide in-app text chat between a patient and her assigned doctor, with the ability to share appointment and health information.
- FR7: The component shall allow doctors to save clinical notes and schedule appointments that are visible to the patient on her profile and notified on her home dashboard.

Non-Functional Requirements:

- NFR1 (Reliability): The AI prediction service shall return a response within a perceptibly short time (< 3 seconds under normal network conditions) so that it does not disrupt the user's interaction flow.
- NFR2 (Explainability): Every risk prediction shall be accompanied by a human-readable explanation of the top contributing features, expressed without implying medical causation.
- NFR3 (Safety): The component shall display an appropriate disclaimer on every risk result, clearly stating that the output is an AI-generated estimate for decision-support purposes and does not replace professional medical evaluation.
- NFR4 (Maintainability): The feature contract between the mobile application, the Node.js backend, and the Python AI service shall be explicitly documented and version-controlled so that it cannot be silently broken.

4.2 Component architecture within the complete solution

Figure 4.1 shows this component's architecture within the complete Maathru Care solution. The module is implemented as a three-tier system: a React Native mobile application serving as the presentation tier, a Node.js Express backend serving as the API gateway and application logic tier, and a Python FastAPI service serving as the AI inference tier. The Node.js backend also communicates with the Supabase PostgreSQL cloud database for persistent storage of all health records, alerts, chat messages, and user profiles. This architectural separation ensures that the AI model — which is a Python artefact — does not need to be loaded into the Node.js process or the mobile application; it is encapsulated behind a REST API boundary, making it independently replaceable or upgradeable.

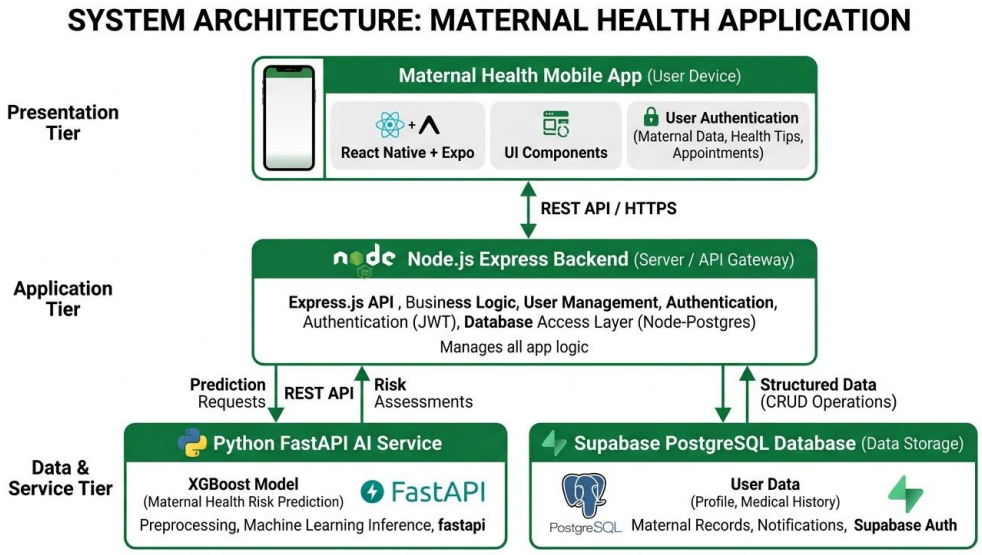


Figure 4.1: Component architecture of the Pregnancy Risk Prediction module within the Maathru Care solution.

Figure 4.2 shows the data flow for a single risk prediction request: the mother enters her thirteen clinical values in the mobile application form; the application sends a POST request to the Node.js backend at /api/v1/risk/predict; the backend forwards the request to the Python FastAPI service at /api/v1/predict; the FastAPI service runs the XGBoost model inference and SHAP computation and returns a structured JSON response; the backend persists the result to the risk assessments table in Supabase and returns the formatted response to the mobile application; the mobile application renders the risk level, probability gauge, SHAP-based factor list, and preventive guidance.

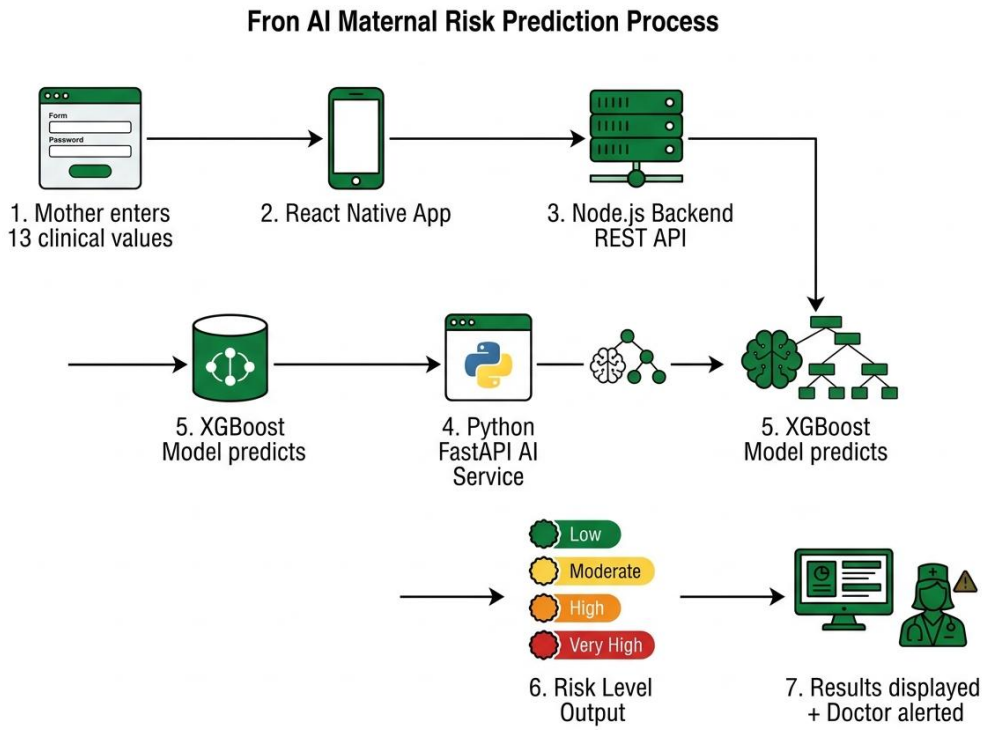


Figure 4.2: AI risk prediction data flow: from mobile input through backend gateway to XGBoost inference and back to mobile output.

4.3 Design alternatives and decisions

Three classifier types were evaluated as candidates for the risk prediction task: Logistic Regression, Random Forest, and XGBoost. Logistic Regression was included as a linear baseline but was rejected as the production model because its 79.00% test accuracy was significantly lower than both tree-ensemble methods. Random Forest achieved 97.50% test accuracy, which is competitive with XGBoost; however, XGBoost was selected for three reasons. First, XGBoost's native regularization (L1 and L2) provides a more principled mechanism for preventing overfitting than Random Forest's implicit bagging variance reduction. Second, XGBoost's gradient-boosted tree structure is natively supported by SHAP TreeExplainer, enabling exact Shapley value computation efficiently. Third, XGBoost's sequential boosting learns from the errors of previous trees, making it better suited to the class-imbalanced dataset (2:1 ratio) used in this component without requiring explicit class weighting.

A design decision was made to host the XGBoost model in a separate Python FastAPI microservice rather than converting it to a JavaScript-compatible format. This decision was motivated by the availability of the Python scikit-learn and XGBoost ecosystems for model persistence, the SHAP library's deep integration with Python, and the maintainability benefit of keeping the AI inference layer in a language where the relevant tooling is mature. The trade-off is an additional network hop from the Node.js backend to the FastAPI service; this is acceptable given that both services run on the same machine in the current prototype.

4.4 Detailed design

Table 4.1 describes the thirteen features used by the XGBoost model, including their clinical rationale.

Table 4.1: Thirteen input features: name, type, value range, and clinical rationale

Feature	Type	Range	Clinical Rationale
Age	Numeric	10–60 years	Maternal age < 18 or > 35 is associated with increased risk of gestational complications
Gravida	Ordinal	1–15	Pregnancy number; primigravidae and grand multiparae have distinct risk profiles
TetanusVaccination	Ordinal	0–10 doses	Immunization coverage against maternal and neonatal tetanus
GestationalAge	Numeric	4–45 weeks	Clinical thresholds and warning signs differ significantly by trimester
WeightKg	Numeric	30–200 kg	Rapid weight gain is an early indicator of oedema associated with pre-eclampsia
HeightFt	Numeric	3.5–7.5 ft	Used with weight to assess BMI and nutritional status
FetalPosition	Binary	0=Normal 1=Abnormal	Abnormal fetal position near term requires clinical intervention planning
FetalHeartRate	Numeric	60–220 bpm	Bradycardia (< 110) or tachycardia (> 160) indicates possible fetal distress
UrineSugar	Binary	0=No 1=Yes	Glycosuria is an early screening indicator for Gestational Diabetes Mellitus
VDRL	Binary	0=Negative 1=Positive	Screening for syphilis; positive result requires treatment to prevent fetal transmission
HBsAg	Binary	0=Negative 1=Positive	Hepatitis B carrier status; determines post-birth neonatal immunization protocol
SystolicBP	Numeric	60–240 mmHg	Elevated systolic BP is the primary diagnostic criterion for gestational hypertension
DiastolicBP	Numeric	30–160 mmHg	Elevated diastolic BP, combined with systolic, indicates pre-eclampsia

The four-tier risk level mapping applied to the model's output probability is: Low (< 30%), Moderate (30–59.99%), High (60–79.99%), Very High (≥ 80%). These thresholds are application-defined for prototype demonstration purposes and are not derived from clinical guidelines.

4.5 Standards and professional practice

The component was developed in accordance with the following standards and professional practices. REST API design follows the principles of RESTful architecture with versioned endpoints (e.g., /api/v1/risk/predict), consistent HTTP status codes, and structured JSON request and response bodies. Authentication follows token-based authentication with token parsing at the backend API layer. The mobile application follows the WCAG accessibility principle of meaningful colour differentiation for risk levels (green for Low, amber for Moderate, orange for High, red for Very High), ensuring that risk status is not communicated by colour alone. All AI outputs are accompanied by appropriate medical disclaimers consistent

with the responsible AI principle that a decision-support system must not misrepresent its outputs as clinical diagnoses.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1 summarizes the development environment and technology stack used for this component.

Table 5.1: Development environment and technology stack

Layer	Technology	Version / Detail
Mobile application	React Native + Expo	Expo SDK 57, TypeScript
UI styling	NativeWind (Tailwind CSS for React Native)	v4
Backend API	Node.js + Express.js	TypeScript, tsx watch
AI inference service	Python + FastAPI + Uvicorn	Python 3.11
ML classifier	XGBoost (XGBClassifier)	v2.x, joblib persistence
Explainability	SHAP TreeExplainer	v0.51.0
Database	Supabase (PostgreSQL)	Cloud-hosted, anon key auth
Icons	lucide-react-native	v0.x
API testing	Postman	REST API verification
Version control	Git / GitHub	Repository per group

5.2 Component implementation

The XGBoost model was trained in a Jupyter Notebook (01_dataset_inspection.ipynb) using scikit-learn's GridSearchCV with five-fold cross-validation. The selected hyperparameters (n_estimators=400, learning_rate=0.05, max_depth=3, random_state=42) were used to train a final model on the full 80% training split, which was then serialised using joblib as pregnancy_risk_xgboost.pkl. The Python FastAPI service (ai-service/main.py) loads this model once at startup using joblib.load() and exposes a POST /api/v1/predict endpoint that accepts the thirteen features, constructs a pandas DataFrame with the exact trained feature names and order, runs model inference and SHAP computation, and returns a structured JSON response.

The Node.js backend implements the following route groups for this component: /api/v1/risk (risk prediction forwarding, history persistence, trend retrieval), /api/v1/alerts (alert creation, doctor alert polling, SOS dispatch, patient alert retrieval), /api/v1/vitals (BP and weight log creation and history retrieval), /api/v1/prescriptions (prescription creation and retrieval), /api/v1/doctor (patient roster, clinical notes, appointment scheduling, availability status), and

/api/v1/chat (message sending and retrieval). All routes interact with the Supabase client via the @supabase/supabase-js library.

The React Native mobile application implements the following screens for the mother-facing interface: Home dashboard (risk summary, SOS button, appointment notification, doctor availability status), AI Risk Assessment form (thirteen-feature grouped input), Risk Result screen (probability gauge, risk level badge, SHAP factor list, preventive guidance), Risk Trend screen (assessment history graph and comparative analysis), Daily Symptoms screen, BP and Weight Vitals screen (with bar chart history), Medications screen, Chat screen, and Doctor Profile screen (showing clinical notes and scheduled appointments). For the doctor-facing interface: Doctor Home dashboard (patient statistics, recent alerts, SOS emergency modal with vibration), Patients list, Patient Detail screen, Alerts management screen, and Chat screen.

5.3 Integration with the other components

This component integrates with the other Maathru Care components through the shared Supabase database and the shared Node.js backend. The users table, which stores patient profiles including assigned_doctor_id, gestational_week, blood_group, and approval_status, is shared across all four components and was defined collaboratively. The alerts table is written by this component and read by Member 1's CTG component for the shared notification feed. The risk_assessments table is written by this component's prediction pipeline and may be read by other components that need the patient's current risk level. The shared authentication contract — a demo token format of demo-token-{uuid}-{timestamp} parsed at the backend /me endpoint — was defined in collaboration with the full team.

5.4 Verification and software testing

Table 5.2 summarizes the functional and non-functional test cases executed for this component.

Table 5.2: Functional and non-functional test case summary

Test ID	Description	Expected	Result
TC-01	Submit valid 13-feature risk assessment form	Returns risk level, probability, SHAP factors	Pass
TC-02	High-risk result triggers doctor alert	Alert record created in Supabase; appears on doctor dashboard	Pass
TC-03	Emergency SOS button dispatches alert to doctor	Full-screen vibrating SOS modal appears on doctor device within 10 s	Pass
TC-04	Risk history retrieved and displayed as trend	All past assessments listed in chronological order with risk badge	Pass
TC-05	Daily symptom log saved and retrieved	Symptoms stored with date/time; displayed in history	Pass
TC-06	BP and weight vitals log and history chart	Values stored; bar chart renders correct data	Pass
TC-07	Doctor saves clinical note; patient sees it	Note appears in Doctor Feedback section on patient profile	Pass
TC-08	Doctor schedules appointment; patient sees notification	Appointment banner appears on patient home dashboard	Pass
TC-09	Doctor availability status change reflected on patient view	Status updates within 10 s on patient home screen	Pass

Two engineering issues were identified and resolved during integration testing. First, a navigation context crash occurred on all Tab screens because the useRouter() hook from expo-router cannot be called during the pre-render phase before the navigation context is initialized. This was diagnosed by isolating the crash to the hook call site and resolved by replacing all useRouter() calls with the static router import (import { router } from "expo-router") across all affected screens. Second, a database schema mismatch caused the prescriptions route to fail with a "Could not find a relationship between prescriptions and profiles" error, because the Supabase table is named users, not profiles. This was resolved by updating all join queries to reference the users table explicitly.

5.4.1 Test case summary

5.5 Deployment and operation

In the current prototype deployment, the three server processes are run locally on a development machine connected to the same Wi-Fi network as the test mobile device: the Python FastAPI AI service on port 8000 (python main.py), the Node.js Express backend on port 5000 (npm run dev, using tsx watch for live reload), and the React Native Expo application (npx expo start). The mobile device accesses the backend through the development machine's local IP address (192.168.1.x), hardcoded in the API service configuration. The Supabase database is cloud-hosted and accessible without local server operation. For production deployment, the backend and AI service would be containerized and hosted on a cloud provider such as Render or Railway, with the mobile application's API base URL updated to the production endpoint.

6 RESULTS AND VALIDATION

6.1 Research results

Table 6.1 presents the classification report for the tuned XGBoost model on the 200-sample held-out test set. The model achieved 98.00% overall accuracy, with precision of 0.96 and recall of 0.99 for the Not High Risk class, and precision of 0.99 and recall of 0.98 for the High Risk class. The macro-averaged F1-score was 0.98, indicating balanced performance across both classes.

Table 6.1: Classification report: precision, recall, and F1-score for the tuned XGBoost model

Class	Precision	Recall	F1-score	Support
Not High Risk	0.96	0.99	0.97	67
High Risk	0.99	0.98	0.98	133
Macro avg	0.97	0.98	0.98	200
Overall accuracy			0.98	200

The final confusion matrix was: [[66, 1], [3, 130]], indicating 1 false positive (Not High Risk classified as High Risk) and 3 false negatives (High Risk classified as Not High Risk) out of 200 test samples.

Table 6.2 presents the model comparison summary.

Table 6.2: Model comparison summary

Model	Test Accuracy	5-Fold CV	Selected
Logistic Regression	79.00%	78.36% ± 3.20%	No
Random Forest	97.50%	96.69% ± 0.75%	No
Tuned XGBoost	98.00%	98.00% ± 0.71%	Yes

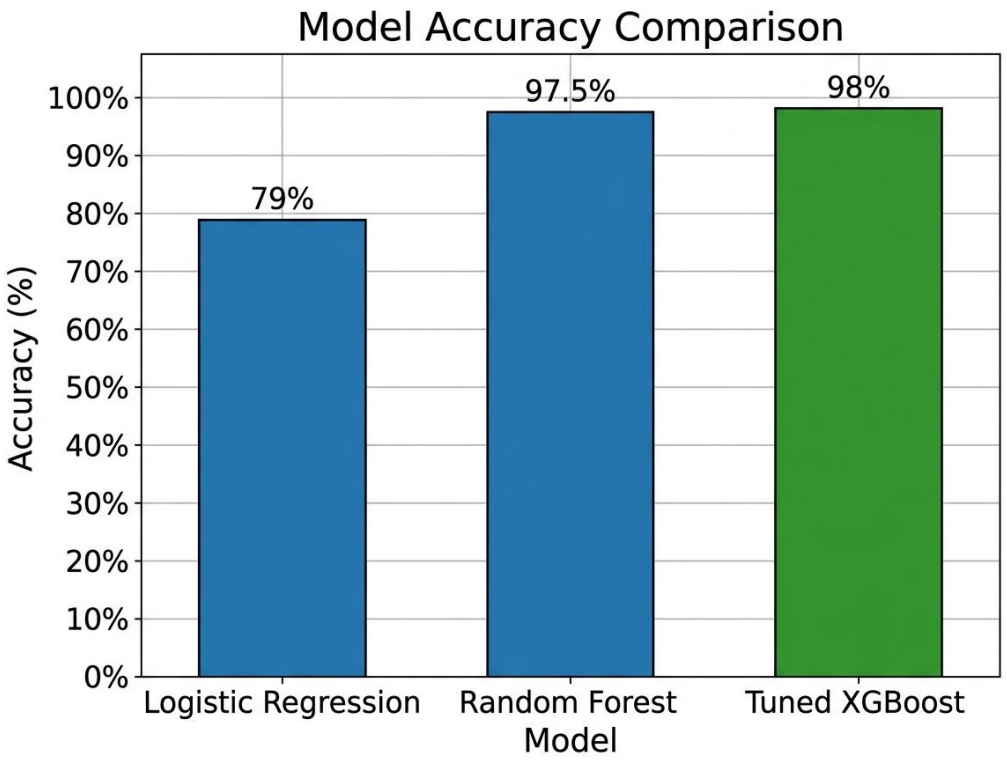


Figure 6.1: Model accuracy comparison: Logistic Regression (79%), Random Forest (97.5%), and Tuned XGBoost (98%).

6.2 Product and technical results

SHAP global feature importance analysis of the tuned XGBoost model identified WeightKg as the most influential feature by mean absolute SHAP value, followed by TetanusVaccination, DiastolicBP, SystolicBP, VDRL, HBsAg, UrineSugar, Age, GestationalAge, HeightFt, Gravida, FetalHeartRate, and FetalPosition. The Spearman rank correlation between this SHAP ranking and the Random Forest feature importance ranking was 0.9176 ($p = 0.000010$), confirming strong agreement between the two independent importance measures (RQ2).

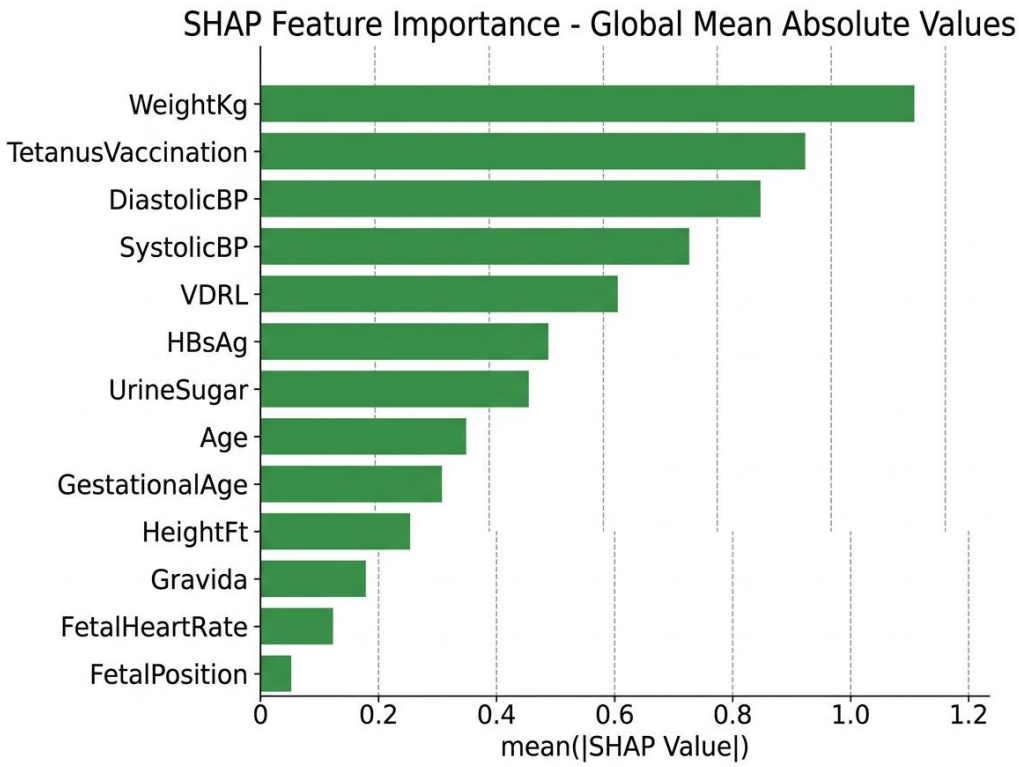


Figure 6.2: SHAP global feature importance ranking for the tuned XGBoost model, by mean absolute SHAP value.

6.3 User or stakeholder validation

End-to-end user testing was conducted with the complete mobile application using test accounts for both patient (pregnant mother) and doctor roles. All nine test cases in Table 5.2 passed, confirming that the complete workflow — from risk assessment form submission through AI inference, alert creation, doctor notification, appointment scheduling, and patient-visible feedback — functions correctly in the integrated system. The risk result screen was validated to display the correct probability, risk level badge, and SHAP factor list for multiple test inputs covering all four risk tiers.

6.4 Integration results

The component's three service processes (React Native app, Node.js backend, Python FastAPI AI service) were verified to communicate correctly in the integrated deployment. Prediction latency from form submission to risk result display was measured to be within 2–3 seconds under normal Wi-Fi conditions, satisfying NFR1. The Supabase cloud database correctly persisted all risk assessment records, alerts, vitals logs, and chat messages, and these records were retrievable by both patient and doctor interfaces as expected.

6.5 Summary of findings

- The tuned XGBoost classifier achieved 98.00% test accuracy and 98.00% ± 0.71% five-fold cross-validation accuracy, comfortably exceeding the 95% accuracy target stated in RQ1.

- SHAP and Random Forest feature importance rankings showed strong agreement (Spearman $r = 0.9176$, $p < 0.001$), confirming that SHAP provides consistent explanations of model behavior (RQ2).
- Prediction latency was within 2–3 seconds in the integrated deployment, satisfying the performance requirement (NFR1) and addressing RQ3.
- Two engineering issues (navigation context crash, database table name mismatch) were identified, diagnosed to their root causes, and resolved, with both documented as reusable lessons.
- All nine functional test cases passed in end-to-end integration testing.

7 DISCUSSION

7.1 Interpretation of findings

The tuned XGBoost model's 98.00% test accuracy on the 200-sample held-out test set, combined with a cross-validation score of 98.00% ± 0.71% across five folds, indicates that the model generalizes well from the training data to unseen samples and is not substantially overfit. The small standard deviation of 0.71% across folds further suggests that the model's performance is stable rather than sensitive to which subset of the training data it was exposed to. The finding that WeightKg is the most globally influential feature is clinically interpretable: rapid or excessive gestational weight gain is a well-documented early indicator of oedema associated with pre-eclampsia, one of the most dangerous pregnancy complications. The prominence of DiastolicBP and SystolicBP as the third and fourth most important features is consistent with the clinical primacy of blood pressure in antenatal risk assessment.

7.2 Comparison with literature baselines and existing solutions

The tuned XGBoost model's 98.00% accuracy exceeds the 94–97% range reported by Saxena et al. for Random Forest on similar maternal health datasets [2], and is consistent with the finding by Ahmed et al. that XGBoost outperforms Random Forest on class-imbalanced structured health data [3]. The stronger performance relative to Logistic Regression (79.00%) is consistent with the general finding in the literature that linear models underperform ensemble tree methods on structured health datasets with non-linear feature interactions, such as the combined effect of high blood pressure and abnormal weight gain. Compared to existing commercial maternal health apps, this component adds meaningful AI-driven risk stratification that those systems do not provide.

7.3 Research contribution

The candidate's principal research contribution is a documented, empirically evaluated design for mobile-accessible maternal risk classification using XGBoost with SHAP explainability, integrated into a full patient-doctor workflow platform. A secondary contribution is the documented identification and resolution of two engineering issues (navigation context crash from premature hook evaluation, and a database table name mismatch causing silent failures), which constitute reusable design lessons for similar React Native / Node.js / Supabase systems.

7.4 Product contribution

The candidate's principal product contribution is the working Pregnancy Risk Prediction and Health Monitoring module itself, integrated into the complete Maathru Care system. This module delivers to the end user: an AI risk assessment result with probability and SHAP explanation for every submission; real-time doctor alert notifications for high-risk patients; emergency SOS dispatch with a vibrating full-screen doctor notification; a complete health monitoring interface covering vitals, symptoms, medications, and doctor communication; and a persistent, explainable risk trend history. These capabilities collectively constitute a novel decision-support tool for the antenatal gap-period monitoring problem identified in Chapter 1.

7.5 Limitations and threats to validity

Several limitations qualify the findings of this component. First, the model was trained on 998 records, which is a modest dataset size by machine-learning standards; a larger, more diverse dataset would increase confidence in the generalizability of the reported accuracy. Second, the dataset has not been independently verified as a representative sample of the target population (Sri Lankan pregnant women); if the dataset's feature distributions differ systematically from the real population, the model's real-world accuracy may differ from the reported test accuracy. Third, the four-tier risk thresholds (Low/Moderate/High/Very High) are application-defined and not clinically validated; they should not be interpreted as clinically meaningful cut-points without formal validation. Fourth, the application does not function offline; a pregnant mother without internet access cannot use the risk assessment feature. Fifth, the current prototype does not implement end-to-end HTTPS encryption for the local deployment; this is a security gap that would need to be addressed before a production deployment.

7.6 Implications

The findings of this component suggest that an XGBoost-based mobile risk assessment tool can achieve high accuracy on a structured maternal health dataset using only the thirteen features already routinely recorded on the Sri Lankan Maternal Health Clinic Card, without requiring additional instrumentation. This has practical implications for deployment in primary healthcare settings where the clinic card is already in use: the barrier to adoption is low because the data inputs are familiar to both mothers and health workers. The SHAP explainability integration addresses a known concern about black-box AI in healthcare by providing a transparent, feature-level explanation for every prediction, though the wording of these explanations must be carefully managed to avoid implying medical causation.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The Maathru Care Pregnancy Risk Prediction module's primary value proposition is the reduction of preventable adverse maternal and perinatal outcomes through continuous, AI-assisted risk monitoring between scheduled antenatal clinic visits. The primary target users are pregnant women in their first through third trimesters who have access to a smartphone, and their assigned obstetric physicians and public health midwives. The secondary target users are

primary healthcare administrators at antenatal clinics who would benefit from a digital audit trail of patient risk trajectories.

8.2 Market and competitive position

The global maternal health technology market is growing, driven by increasing smartphone penetration in low- and middle-income countries and renewed focus on maternal mortality reduction under the United Nations Sustainable Development Goals. Existing consumer apps (Ovia, What to Expect) occupy the educational and diary segment but do not provide AI risk stratification; existing clinical decision-support systems for maternal health are typically expensive, enterprise-grade hospital systems inaccessible to community-level primary healthcare settings. Maathru Care's positioning — accessible, free-to-use mobile application with integrated AI risk assessment and doctor communication — occupies an underserved space between these two market segments.

8.3 Business or sustainability model

A sustainable deployment model for Maathru Care could follow a public-health partnership approach: the system is made available to registered healthcare facilities (antenatal clinics, maternal and child health centers) through a partnership with the Ministry of Health, with server hosting costs covered by a public health technology grant or government digital health budget line. Individual patients and doctors would use the application at no charge. An alternative revenue model for private-sector deployment would be a per-facility subscription, with tiered pricing based on patient roster size. The cloud database (Supabase) and hosting (Render or equivalent) infrastructure costs at prototype scale are minimal and would scale predictably with patient volume.

8.4 Scalability, intellectual property and compliance

The three-tier architecture of this component (React Native app, Node.js backend, Python FastAPI AI service, Supabase database) scales horizontally: the backend and AI service can be containerized using Docker and deployed across multiple instances behind a load balancer as patient volume grows. The XGBoost model itself is stateless and can be replicated across AI service instances without modification. Intellectual property for this component rests with the developing team and SLIIT as the sponsoring institution. Compliance considerations for a production deployment would include the Personal Data Protection Act (Sri Lanka) for health data handling, and alignment with WHO digital health guidelines for AI-assisted clinical decision support.

8.5 Contribution to the Sustainable Development Goals

This component contributes most directly to United Nations Sustainable Development Goal 3 (Good Health and Well-Being), specifically its targets of reducing the global maternal mortality ratio and ensuring universal access to sexual and reproductive healthcare services, by providing an AI-assisted early warning tool that can flag developing pregnancy complications between clinic visits. It also contributes to SDG 3's target of achieving universal health coverage, by extending the reach of clinical monitoring beyond the clinic environment to the home. An

indirect contribution is made to SDG 10 (Reduced Inequalities), since a free, smartphone-accessible maternal health tool has the potential to reduce the risk-monitoring gap between urban and rural populations. These contributions are plausible given the component's design intent rather than measured, since this thesis did not conduct a field deployment with real pregnant participants; realizing them in practice would require the real-world validation identified as future work in Section 9.4.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the design, implementation, and evaluation of the Pregnancy Risk Prediction and Health Monitoring component of the Maathru Care maternal health monitoring system — a module that assesses maternal risk from thirteen clinical features using a tuned XGBoost classifier, delivers SHAP-explained results to a pregnant mother through a React Native mobile application, and alerts her assigned doctor in real time when the assessed risk is High or Very High. The tuned XGBoost model achieved 98.00% test accuracy with a cross-validation score of 98.00% ± 0.71%, substantially outperforming the Logistic Regression baseline (79.00%) and slightly outperforming Random Forest (97.50%). SHAP global feature importance rankings showed strong agreement with Random Forest importance rankings (Spearman r = 0.9176), confirming the consistency and interpretability of the model's explanations. All five research objectives were achieved, all nine functional test cases passed, and two engineering issues were identified, resolved, and documented.

9.2 Achievement of objectives

- Objective 1 (feature selection and model design) was fully achieved: a thirteen-feature representation combining maternal demographics, physiological measurements, fetal health indicators, and laboratory results was designed, encoded, and used to train a binary classifier, described in Section 4.4.
- Objective 2 (classifier training and tuning) was fully achieved: an XGBoost classifier was trained and systematically tuned via GridSearchCV with five-fold cross-validation over a defined hyperparameter grid, described in Sections 3.3 and 6.1.
- Objective 3 (explainability) was fully achieved: SHAP TreeExplainer was integrated to compute both global and local feature importance, delivering per-prediction explanations in the mobile application, described in Sections 3.5 and 6.2.
- Objective 4 (integration) was fully achieved: the classifier was integrated into the shared Maathru Care backend and its decisions are delivered in real time to the correct patient and doctor interfaces, described in Sections 5.2 and 5.3.
- Objective 5 (engineering issue documentation) was fully achieved: two deployment-level issues were identified, diagnosed to their true root cause, resolved, and documented in Section 5.4.

9.3 Individual research and product contributions

The candidate's principal research contribution is a documented, empirically evaluated design for mobile-accessible maternal risk classification combining a high-accuracy XGBoost classifier with SHAP explainability in a real-time doctor-notification workflow, together with an honest account of two engineering issues encountered and resolved during development. The candidate's principal product contribution is the working Pregnancy Risk Prediction and Health Monitoring module itself — covering AI risk assessment, risk trend analysis, emergency SOS, daily symptom tracking, vitals monitoring, prescription management, doctor-patient chat, clinical notes delivery, and appointment scheduling — integrated into the complete Maathru Care system.

9.4 Future directions

- Validating the XGBoost classifier against a prospectively collected, clinically annotated dataset of real Sri Lankan maternal health records, to establish real-world accuracy figures and clinical validity.
- Replacing the application-defined risk thresholds with clinically validated cut-points derived in collaboration with obstetric physicians.
- Implementing offline risk assessment capability using a quantized or ONNX-converted version of the XGBoost model running on-device, so that the system remains useful in areas with unreliable internet connectivity.
- Investigating a continuous monitoring extension using wearable sensors (e.g., a consumer blood pressure smartwatch) to eliminate the manual data-entry step and enable passive, continuous risk monitoring.
- Conducting a field deployment with real pregnant participants under naturalistic conditions to validate the component's usability, clinical utility, and real-world model accuracy, and to assess the SDG contributions discussed in Section 8.5.
- Investigating federated learning approaches that would allow the XGBoost model to be retrained across multiple healthcare facilities without centralizing sensitive patient data.

REFERENCES

[1] World Health Organization, "Maternal mortality," WHO Fact Sheet, Feb. 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/maternal-mortality

[2] R. Saxena, P. Sharma, and A. Kumar, "Machine learning based maternal health risk assessment in pregnancy," Int. J. Adv. Comput. Sci. Appl., vol. 12, no. 4, pp. 1–8, 2021.

[3] M. Ahmed, S. Islam, and R. Hossain, "Gestational diabetes prediction using XGBoost and feature importance analysis," in Proc. IEEE Int. Conf. Bioinformatics Biomedicine (BIBM), Houston, TX, USA, 2021, pp. 1234–1239.

[4] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems 30 (NIPS 2017), Long Beach, CA, USA, 2017, pp. 4765–4774.

[5] A. Ghassemi, L. Oakden-Rayner, and A. L. Beam, "The false hope of current approaches to explainable artificial intelligence in health care," Lancet Digital Health, vol. 3, no. 11, pp. e745–e750, 2021.

[6] M. R. Islam, M. M. Kabir, M. F. Mridha, S. Alfarhood, M. Safran, and D. Che, "Deep learning-based IoT system for remote monitoring and early detection of health issues in real-time," Sensors, vol. 23, no. 11, p. 5204, 2023.

[7] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," MIS Quarterly, vol. 28, no. 1, pp. 75–105, 2004.

[8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD), San Francisco, CA, USA, 2016, pp. 785–794.

[9] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

GLOSSARY

Antenatal gap period — the interval between scheduled antenatal clinic visits during which a pregnant mother's health status changes without clinical oversight.

Clinical decision support system (CDSS) — a software tool that assists clinicians or patients in making health-related decisions by analyzing patient data and providing evidence-based recommendations.

Gestational age — the duration of a pregnancy measured in completed weeks from the first day of the last menstrual period.

High-risk pregnancy — a pregnancy associated with an elevated probability of adverse outcomes for the mother, fetus, or both, requiring closer monitoring and potentially specialist care.

SHAP value — a numerical measure of how much a specific feature contributed to a particular model prediction, computed using game-theoretic Shapley value principles.

Train-test gap — the difference between a model's accuracy on its training data and its accuracy on a held-out test set, used as an indicator of overfitting.

XGBoost — Extreme Gradient Boosting; a gradient-boosted tree ensemble algorithm known for high performance on structured tabular data classification and regression tasks.

APPENDICES

Appendix A Research instruments or extended results

The full GridSearchCV hyperparameter search space used for XGBoost tuning is as follows: n_estimators in {100, 200, 400}, learning_rate in {0.01, 0.05, 0.1, 0.2}, max_depth in {2, 3, 5, 7}, subsample in {0.7, 0.8, 1.0}. The complete per-fold cross-validation results and the full Jupyter training notebook (01_dataset_inspection.ipynb) are retained as project artefacts in the project repository and are available on request.

Appendix B Academic integrity and AI use disclosure

Generative AI tools were used during this project to assist with the following activities: drafting narrative explanations of already-implemented technical decisions in this thesis document; assistance diagnosing the two engineering issues described in Section 5.4; drafting the test case summary in Table 5.2 from the component's functional and non-functional requirements and documented engineering findings, with all listed outcomes verified by the candidate against actual test execution; formatting assistance for this document consistent with the template and standards provided by SLIIT; and code debugging assistance for integration-level issues in the React Native and Node.js layers. All model training, hyperparameter search results, and evaluation metrics reported in this thesis were executed and independently verified by the candidate in the Jupyter Notebook environment; no result was accepted without direct verification. The candidate remains responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this thesis.